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Hoist and supplemental support system

Abstract

A support assembly comprises a bracket, a blocking member, and a spring. The blocking member is adapted to slidingly move within a through opening of a support member. The blocking member is adapted to be selectively moved from a supporting position to a non-supporting position. The spring is adapted to bias the blocking member towards the non-supporting position. In certain embodiments, the bracket is adapted to retain the blocking member at least partially within the through opening when the bracket is mounted to a hoist. In certain embodiments, the blocking member comprises a handle comprising a pin axially aligned within a compression spring and the bracket comprises a notch adapted to receive the handle.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3317004	12/1966	Harrison, Jr.	254/91	B66F 7/0641
3811338	12/1973	Federspiel	74/527	B60P 1/283
2005/0262908	12/2004	Parker	70/177	G05G 1/082

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Background/Summary

(1) This application is a continuation application of U.S. application Ser. No. 17/009,201, filed Sep. 1, 2020, the contents of which are incorporated herein by reference and of which Applicant claims the benefit.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to power hoists and more specifically to a low or minimum profile hoist and supplemental support system adapted to prevent the hoist from collapsing.

2. Description of the Prior Art

(2) Lifts or hoists are used in a variety of applications, including, for, example, use in dump truck bodies. There are a number of hoist types known in the art, including scissor type hoists and telescopic hoists. The scissor type hoist provides great stability but is less efficient than the telescopic hoist. The telescopic hoist, though more efficient and, in many cases, more powerful, is less stable than the scissor type hoist, particularly when lifting uneven loads. Combination hoists were developed to take advantage of the stability of scissor variety hoists and the strength of the telescopic hoists. Combination hoists generally comprise an upper frame assembly pivotally coupled to a lower frame assembly, and one or more hydraulic arms extending from a base to the upper arm. Extension of the hydraulic arms results in the raising of the lower and upper frames, which in turn causes the raising of the bed of the dump trailer. Conversely, lowering the lower and upper frames involves retracting the hydraulic arm or arms.

(3) As long as sufficient hydraulic pressure is maintained within the hydraulic system, the dump

bed or other object being supported will remain in an extended position. However, hydraulic systems can lose hydraulic pressure either intentionally or accidentally. When hydraulic pressure is released from the lifting portion of the hoist, the lower and upper frames will lower and begin to collapse. A sudden release of such hydraulic pressure can result in a sudden collapse of the hoist that can result in severe injury or death.

(4) What is needed is a simple and reliable system that will prevent a hoist from collapsing upon the release of hydraulic pressure.

SUMMARY OF THE INVENTION

(5) The hoist generally comprises a base, lower arms, upper arms, hydraulic cylinder assembly, upper arms crossmember, and one or supplemental support systems. The base is adapted to be coupled to a frame of a vehicle such as a dump truck. Pivotally coupled to the base are lower arms and the hydraulic cylinder assembly. In the preferred embodiment, a base axle is threaded through openings in lower arm proximal ends and a hydraulic cylinder lower retainer. Pivotally coupled to distal ends of the lower arms are respective upper arms. The upper arms of the preferred embodiment are stabilized by upper arms crossmember which spans across upper ends of the upper arms and bushing which spans between lower ends of the upper arms. An arm axle threads through lower arm upper through openings and upper arm lower through openings.

(6) The one or more supplemental support systems each comprise a bracket, a blocking member/pin, a spring and a pin handle. The pin is adapted to be slidably positioned within a pin through opening in the lower arm, and in some embodiments, through a central bracket through opening. The bracket of the preferred embodiment is adapted to extend orthogonally from an outside portion of the lower arm and comprises an angular “U” configuration. In other embodiments, the bracket comprises a closed rectangular configuration. In angular “U” configuration embodiments, legs of the bracket are directly connected to the outside portion of the lower arm. In closed rectangular configuration embodiments, a distal short side of the rectangle is closed and comprises the central bracket through opening through which the pin is adapted to slidably move. Thus, the supplemental support system is adapted to be either built into the hoist or serve as an accessory that can be added to a conventional hoist as an accessory.

(7) The pin handle extends orthogonally from an outside end of the pin. The spring of the preferred embodiment is an open coil helical compression spring. The spring surrounds a portion of the pin. With this configuration, legs of the bracket are coupled to the outside portion of the lower arm such that the bracket legs straddle the pin positioned in the pin through opening and the spring. The bracket further comprises a notch adapted to receive the pin handle when the pin is in a supporting position. The pin is adapted to slide within the pin through opening such that, when the pin is in the supporting position, a distal end of pin is positioned beneath upper arm.

(8) The hydraulic cylinder assembly comprises a barrel and a piston rod, the piston rod being adapted to extend and retract. The piston rod is structured and arranged to be removably coupled with a rod receiver portion. A rod receiver crossmember is pivotally coupled to upper arms via rod receiver crossmember axle such that the rod receiver portion is axially aligned with the rod receiver crossmember when coupled with the rod receiver crossmember.

(9) In preferred embodiments, the upper arm comprises a pad. This pad is positioned between the distal end of pin and upper arm when the pin is in the supporting position.

(10) In preferred embodiments, the hoist comprises tubular lower arms and upper arms.

(11) In preferred embodiments the supplemental support system is coupled to an upper leg such that the pin, in the supporting position rests against the lower leg.

(12) In preferred embodiments, the hoist and supplemental support are formed from heavy duty steel adapted to withstand heavy loads.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front and right-side isometric view of the hoist in a retracted condition, in accordance with a preferred embodiment.
- (2) FIG. 2 is a front and right-side isometric view of the hoist in an extended condition, in accordance with a preferred embodiment.
- (3) FIG. 3 is a front and right-side isometric view of the supplemental support system portions of the hoist of FIGS. 1 and 2 with the pins in the supporting position.
- (4) FIG. 4 is a front and left-side isometric view of the supplemental support system portions of the hoist of FIGS. 1 and 2 with the pins in the non-supporting position.
- (5) FIG. 5 is close up view of the right pin latch assembly of FIGS. 1-4.
- (6) FIG. 6 is an isometric view of the handle of the supplemental support system in accordance with a preferred embodiment.
- (7) FIG. 7 is an isometric view of the supplemental support system in accordance with a preferred embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENT

- (8) Referring to FIGS. 1-5, & 7, there is shown the hoist **12** and supplemental support system **14** in accordance with preferred embodiments. As used herein, the terms “a” or “an” shall mean one or more than one. The term “plurality” shall mean two or more than two. The term “another” is defined as a second or more. The terms “including” and/or “having” are open ended (e.g., comprising). The term “or” as used herein is to be interpreted as inclusive or meaning any one or any combination. Therefore, “A, B or C” means “any of the following: A; B; C; A and B; A and C; B and C; A, B and C”. An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.
- (9) Reference throughout this document to “one embodiment,” “certain embodiments,” “an embodiment,” or similar term means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, the appearances of such phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner on one or more embodiments without limitation. The detailed description illustrates by way of example, not by way of limitation, the principles of the invention. This description will clearly enable one skilled in the art to make and use the invention, and describes several embodiments, adaptations, variations, alternatives, and uses of the invention, including what is presently believed to be the best mode of carrying out the invention.
- (10) Referring to FIGS. 1 and 2, the hoist **12** generally comprises a base **18**, lower arms **20**, upper arms **22**, hydraulic cylinder assembly **24**, upper arms crossmember **26**, and one or more supplemental support systems **14**. The base **18** is adapted to be mounted to a frame **16**, such as the frame **16** of a vehicle comprising, for example, a rectangular frame **16** configuration and one or more frame **16** crossmembers **56**.
- (11) Pivotally coupled to the base **18** are lower arms **20**, **20** and the hydraulic cylinder assembly **24**. In the preferred embodiment, a base axle **32** is threaded through openings (not shown) in lower arm proximal ends **36** and a hydraulic cylinder lower retainer **34**. Pivotally coupled to distal ends **38** of the lower arms **20**, **20** are respective upper arms **22**, **22**. The upper arms **22**, **22** of the preferred embodiment are stabilized by upper arms crossmember **26** which spans across upper ends of the upper arms **22**, **22** and further stabilized by a bushing **66** which spans between lower ends of the upper arms **22**, **22**. An arm axle **30** threads through lower arm upper through openings **68** and upper arm lower through openings **70**. The crossmember comprises a crossmember through opening **72** adapted to receive a crossmember axle (not shown) that can be mounted between frame members of a truck bed (not shown).

(12) In the preferred embodiment, each lower arm **20, 20** comprises an angle **70** such that each lower arm **20, 20** is not straight. Rather, the distal end **38** of each lower arm **20, 20** extends downward relative to longitudinal axis of the lower arm **20**. Each lower arm **20, 20** further comprises a tapering configuration such that each lower arm **20, 20** is narrower in side profile nearer the proximal end **36** than more distal to the proximal end **36**. Each upper arm **22, 22** of the preferred embodiment comprises a tapering configuration such that each upper arm **22, 22** is narrower in side profile nearer the upper arms crossmember **26** than farther from the upper arms crossmember **26**.

(13) As best seen in FIGS. 3-5, and 7, the one or more supplemental support systems **14** each comprise bracket **40**, blocking member **44** (hereinafter referred to as pin **44**), spring **46** and pin handle **42**. The pin **44** is adapted to be slidably positioned within a pin through opening **48** in the lower arm **20**. Referring to FIG. 6, the handle **42** of certain embodiments is a handle **42** comprising slot **78** adapted to receive a shim (not shown) adapted to maintain the handle **42** in position within the pin **44**. The bracket **40** of the preferred embodiment is adapted to extend orthogonally from an outside portion of the lower arm **20, 20** and comprises a “U” bracket **40** comprising an angular “U” configuration. In other embodiments, the bracket **40** comprises a closed rectangular configuration. In angular “U” configuration embodiments, legs **50, 51** of the bracket **40** are directly connected to an outside portion of the lower arm **20**. In closed rectangular configuration embodiments of the bracket **40**, a distal short side of the rectangle is closed and comprises a central bracket through opening through which the pin **44** is adapted to slidably move. Thus, the supplemental support system **14** is adapted to be either built into the hoist **12** or serve as an accessory that can be added to a conventional hoist.

(14) The pin handle **42** extends orthogonally from an outside end portion of the pin **44**. In preferred embodiments, the pin handle **42** extends through opposite sides of the pin **44**. With this configuration, the spring **46** is retained in general axial alignment with the pin **44**. In alternative embodiments, two pin handles **42, 42** can be arranged such that no through pin opening is necessary.

(15) The spring **46** of the preferred embodiment is an open coil helical compression spring **46**. In some embodiments, the spring **46** may have a torsion component. The spring **46** surrounds a portion of pin **44**. The spring **46** biases the pin **44** towards an end portion **74** of bracket **40**. The torsion attributes of the spring **46** of certain embodiments rotationally biases the pin handle **42** upwards toward a bracket leg **51** comprising a notch **76**.

(16) The end portion **74** and pin through opening **48** retain the pin **44** in axial alignment with the pin through opening **48**. With this configuration, the legs **50, 51** of the bracket **40** are coupled to the outside portion of the lower arm **20** such that the bracket legs **50, 51** straddle the pin **44** positioned in the pin through opening **48** and spring **46**.

(17) The notch **76** of the bracket **40** is adapted to receive the pin handle **42** when the pin **44** is in a supporting position, as shown, for example, in FIG. 3. The pin **44** is adapted to slide within pin through opening **48** such that, when the pin **44** is in the supporting position a distal end **62** of pin **44** is positioned beneath upper arm **22** and when the pin **44** is in a non-supporting position, as shown, for example, in FIG. 4, the distal end **62** of the pin **44** is adjacent to the lower arm **20** and out of a path of downward movement of the upper arm **22**.

(18) In preferred embodiments, the upper arm **22** comprises a pad **64**. This pad **64** is positioned between the distal end **62** of pin **44** and upper arm **22** when the pin **44** is in the supporting position.

(19) The hydraulic cylinder assembly **24** comprises a barrel **66** and a piston rod **52**, the piston rod **52** being adapted to extend and retract in a manner well known in the art. The piston rod **52** is structured and arranged to be removably coupled with rod receiver portion **54** (the piston rod **52** and rod receiver portion **54** shown in an uncoupled condition in FIGS. 2-4). Rod receiver crossmember **58** is pivotally coupled to upper arms **22, 22** via rod receiver crossmember axle **60** (FIG. 3) such that the rod receiver portion **54** is axially aligned with the rod receiver crossmember

58 when coupled with the rod receiver crossmember **58**.

(20) Although a hydraulic cylinder assembly **24** is described herein as the assembly used to move the hoist **12** to and between the positions described herein, other lifting assemblies may be used. For example, pneumatic systems well known in the art can be substituted for the elements of the hydraulic cylinder assembly **24** and used to cause the upper and lower arms **22, 22, 20, 20** to be moved.

(21) The hoist **12** comprises a retracted condition, as shown, for example, in FIG. **1** and an extended condition, as shown, for example in FIG. **2**. In the retracted condition, the lower arms **20, 20** are positioned such that the lower arm distal ends **38** are adjacent to an inside portion of the frame **16** and the upper arms crossmember **26** is adjacent to the base **18**. In the extended condition, the lower arms **20, 20** are positioned such that the lower arm distal ends **38** are above the frame **16** and the upper arms crossmember **26** is raised well above the base **18**.

(22) Although the drawings depict several components of the hoist **12** as being angular, the hoist **12** need not comprise angular components. For example, the base **18**, lower arms **20, 20** and upper arms **22, 22** can comprise tubular elements.

(23) Although in the preferred embodiment, the supplemental support systems **14** are coupled to respective lower legs **20, 20**, the supplemental support system **14** need not be so coupled. Rather, in other embodiments, the supplemental support system **14** may be coupled to one or both upper legs **22, 22** such that the pin **44**, in the supporting position rests against the lower leg **20, 20**.

(24) In preferred embodiments, the hoist **12** and supplemental support **14** are formed from heavy duty steel adapted to withstand heavy loads. However, the hoist **12** can be form from any suitable material known in the art and need not be constructed from steel.

(25) The use of the hoist **12** and supplemental support systems **14** will now be discussed. The hoist **12** is adapted to be mounted to a truck frame **16** and truck bed. In preferred embodiments, the base **18** is mounted to the truck frame **16** and the upper arms crossmember **26** is pivotally mounted to a crossmember axle spanning between support members of the truck bed. The hydraulic cylinder assembly **24** is operatively connected to a source of hydraulic fluid and a pump structured and arranged to move such fluid into and out of the barrel **66** and otherwise activate the hoist **12** as required by a user.

(26) With the hoist **12** in the retracted position such that the truck bed is generally parallel to the truck frame **16**, the user, by activating the hydraulic fluid pump, can cause fluid to enter the barrel of the hydraulic cylinder assembly **24**. Such entry of fluid causes the piston rod **52** to extend which, in turn, causes the hoist **12** to move from the retracted condition depicted in FIG. **1** to the extended condition depicted in FIG. **2**. This movement causes the truck bed to move upward so as to permit, for example, the user to dump contents of the truck bed.

(27) With the hoist **12** in such extended condition, the user may then, grasp the pin handle **42** of each spring assisted pin support system **14** and move the pins **44** from the non-supporting positions depicted in FIG. **4** to the supporting positions of FIG. **3**. The pins **44** are retained in place by the pin handle **38** being positioned within the notch **76**. With the pin **44** in such supporting position, the upper arm **22** is prevented from moving downward past the pin **44** to, for example, the retracted position. Therefore, should the hydraulic cylinder assembly **24** be inadvertently activated so as to attempt to lower the hoist **12** or if hydraulic pressure were to be otherwise lost, the lower arms **20, 20** and upper arms **22, 22**, together with the truck bed, would remain in the extended condition.

(28) The procedure for lowering the hoist **12** from the extended position to the retracted position is generally the reverse of that described above. To do so, the user, by grasping the pin handle **42** of each supplemental support system **14** and removing the pin handles **42** from each notch **76**, moves the pins **44** from the supporting positions depicted in FIG. **3** to the non-supporting positions of FIG. **4**. As the spring **46** biases the pin **44** towards the non-supporting position when the pin handle **42** is removed from the notch **76**, the user can easily return the pins **44** to their non-supporting positions. The user then, by activating the hydraulic fluid pump, can cause fluid to exit the barrel **66** of the

hydraulic cylinder assembly **24**. Such exit of fluid causes the piston rod **52** to retract which in turn causes the hoist **12** to move from the extended condition depicted in FIG. **2** to the retracted condition depicted in FIG. **1**. This movement causes the truck bed to move downward so as to permit, for example, the truck to be driven safely.

(29) The foregoing disclosure and showings made in the drawings are merely illustrative of the principles of this invention and are not to be interpreted in a limiting sense. While there has been illustrated and described what is, at present, considered to be a preferred embodiment of the present invention, it will be understood by those skilled in the art that various changes and modifications may be made, and equivalents may be substituted for elements thereof without departing from the true scope of the invention. Therefore, it is intended that this invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out the invention, but that the invention will include all embodiments falling within the scope of this disclosure and the claims.

Claims

1. A support assembly comprising: a bracket, a blocking member, and a spring, the bracket being adapted to be fixed to a support member comprising a through opening; the through opening comprising first and second through opening ends, such that when the bracket is fixed to the support member, the second through opening end is nearer to the bracket than the first through opening end; the blocking member comprising first and second blocking member ends; the bracket comprising first and second legs and a bracket end portion, the bracket end portion spanning between the first and second legs; when fixed to the support member, the bracket being adapted to: retain the blocking member, at least partially, within the through opening such that: the first end of the blocking member is positioned within the through opening or beyond the first through opening end; the second end of the blocking member is positioned within a space defined by the bracket legs and the bracket end portion; the bracket end portion prevents any portion of the second end of the blocking member to move beyond the bracket end portion; and prevent the second end of the blocking member from exiting the space defined by the bracket legs and the bracket end portion; the blocking member structured and arranged to be axially aligned with the through opening when the bracket is fixed to the support member such that a sliding portion of the blocking member is adapted to slidably move within the through opening while the second end of the blocking member is retained within the space defined by the bracket legs and the bracket end portion; the blocking member being adapted to be selectively moved from a supporting position to a non-supporting position.
2. The support assembly of claim 1, wherein the blocking member is a pin.
3. The support assembly of claim 2, wherein the spring comprises a helical compression spring and wherein the spring surrounds the pin such that the spring and pin are axially aligned.
4. The support assembly of claim 3, wherein the spring is adapted to axially bias the blocking member towards the bracket end portion such that the blocking member is biased towards the non-supporting position.
5. The support assembly of claim 2, further comprising a handle coupled to the pin.
6. The support assembly of claim 5, wherein at least one of the bracket legs comprises a notch adapted to receive the handle.
7. The support assembly of claim 6, wherein the blocking member is in the supporting position when the handle is positioned within the notch.
8. The support assembly of claim 1, wherein the blocking member is adapted to support a portion of a hoist when the blocking member is in the supporting position.
9. The support assembly of claim 1, wherein, when the blocking member is slidably moved within the through opening, the sliding portion of the blocking member is capable of axial rotation about a

longitudinal axis of the blocking member.

10. A support assembly comprising: a bracket, a blocking member, a spring, and a support member, the support member comprising a through opening, the bracket being fixed to the support member; the through opening comprising first and second through opening ends, such that when the bracket is fixed to the support member, the second through opening end is nearer to the bracket than the first through opening end; the blocking member comprising first and second blocking member ends; the bracket comprising first and second legs and a bracket end portion, the bracket end portion spanning between the first and second legs; the bracket being adapted to: retain the blocking member, at least partially within the through opening such that: the first end of the blocking member is positioned within the through opening or beyond the first through opening end; the second end of the blocking member is positioned within a space defined by the bracket legs and the bracket end portion; the bracket end portion prevents any portion of the second end of the blocking member to move beyond the bracket end portion; and prevent the second end of the blocking member from exiting the space defined by the bracket legs and the bracket end portion; the blocking member being axially aligned with the through opening such that a sliding portion of the blocking member is adapted to slidingly move within the through opening while the second end of the blocking member is retained within the space defined by the bracket legs and the bracket end portion; the blocking member being adapted to be selectively moved from a supporting position to a non-supporting position.

11. The support assembly of claim 10, wherein the blocking member is a pin.

12. The support assembly of claim 11, wherein the spring comprises a helical compression spring and wherein the spring surrounds the pin such that the spring and pin are axially aligned.

13. The support assembly of claim 12, wherein the spring is adapted to axially bias the blocking member towards the bracket end portion such that the blocking member is biased towards non-supporting position.

14. The support assembly of claim 12, further comprising a handle coupled to the pin.

15. The support assembly of claim 14, wherein at least one of the bracket legs comprises a notch adapted to receive the handle.

16. The support assembly of claim 15, wherein the blocking member is in the supporting position when the handle is positioned within the notch.

17. The support assembly of claim 10, wherein the support member is a lower arm of a hoist.

18. The support assembly of claim 17, wherein the blocking member supports an upper arm of the hoist when the blocking member is in the supporting position.

19. The support assembly of claim 10, wherein, when the blocking member is slidingly moved within the through opening, the sliding portion of the blocking member is capable of axial rotation about a longitudinal axis of the blocking member.
